Xi Lin

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Summary

Mathematics graduate (Imperial College London) and Master of IT student at UTS (WAM 84.29), focused on retrieval and LLM applications: building RAG systems, fine-tuning retrieval encoders, and evaluating them with controlled experiments. Seeking an AI / machine learning engineering internship for Summer 2026–27.

Education

University of Technology Sydney Sydney, Australia
Master of Information Technology — WAM 84.29 *Jul 2025 – Jun 2027 (expected)*

Imperial College London London, UK
B.Sc. in Mathematics *Sep 2020 – Jun 2024*

Projects

StudyMate — RAG Study Agent with Live Public Demo
Personal Project — GitHub — Live Demo *2026*

- Built and publicly deployed a RAG study agent over lecture notes (LangChain, ChromaDB, OpenAI API, FastAPI) with source-cited answers and an auto-generated, auto-graded quiz mode.
- Built a retrieval evaluation pipeline comparing dense, BM25 and hybrid retrieval; dense retrieval reached **96% Hit@4** on a 700-chunk corpus (LLM-generated queries), with error analysis of why equal-weight hybrid fusion underperformed.
- Added incremental re-indexing via content hashing, and made the public demo safe with isolated per-visitor sessions and rate limiting.

Retrieval Encoder Fine-Tuning and Sample Selection
UTS Research Project — Supervisor: Dr Xianzhi Wang *Aug 2026 – Present*

- Built a contrastive fine-tuning pipeline for MiniLM that mines pseudo-positive document pairs by clustering and nearest-neighbour matching; evaluated on BEIR NFCorpus and SciFact.
- Compared four sample-selection strategies, including split-conformal filtering, across three seeds and two training budgets with bootstrap intervals and Holm correction. Fixed-threshold selection beat size-matched random controls by up to **0.012 nDCG@10** on NFCorpus but did not consistently replicate on SciFact.

Hard Negatives in Dense Retrieval: A Controlled Study
UTS Course Project — Advanced Data Analytics Algorithms, Machine Learning *Sep 2026*

- Implemented masked InfoNCE and the training loop from scratch in PyTorch to test whether BM25-mined hard negatives help fine-tune MiniLM on BEIR FiQA (57k documents; three seeds; sealed test set).
- Random negatives raised test nDCG@10 from 0.369 to **0.397**; hard negatives scored 0.016 lower. A blind review showed hard negatives were far more often on-topic (45% vs. 5%) yet rarely mislabelled, revising my initial hypothesis.

Experience

National Supercomputing Center Wuxi Wuxi, China
Research Intern, Intelligent Computing Department *Oct 2024 – Jan 2025*

- Adapted and ran a custom PyTorch word2vec implementation (no gensim) over a multi-hundred-GB JSON corpus of patent records, producing document representations for a downstream patent-valuation model that scored filings by similarity to patents from the surrounding decade.
- Made training tractable on a single node of a remote NVIDIA GPU cluster by chunking the corpus at the file level and streaming it through the pipeline, keeping memory bounded across ~12-hour full training runs (SSH + VS Code Remote).

Wuxi Haichuang Investment (Private Equity, Series A to pre-IPO) Wuxi, China
Research Assistant *Jun 2023 – Sep 2023*

- Conducted industry research for deal evaluation in semiconductors and pharmaceuticals, including on-site interviews with founders and executives of prospective portfolio companies, and synthesised findings into research reports and decks for the investment team.

Skills

Programming: Python, SQL
ML & Deep Learning: PyTorch model training, contrastive learning & embeddings, retrieval evaluation, RAG & vector search, statistical learning
Frameworks & Tools: PyTorch, Hugging Face Transformers, Sentence-Transformers, scikit-learn, LangChain, ChromaDB, FastAPI, Git, Linux/GPU clusters
Mathematics: Linear Algebra, Probability & Statistics, Optimization, Time Series Analysis
Languages: Chinese (native), English (fluent)

Competitions & Honors

- Dean's List, University of Technology Sydney, Autumn 2026.
- Mathematical Contest in Modeling (MCM), 2021 — Meritorious Award (team).